QUANTUM CRYPTOGRAPHY (BB84 PROTOCOL) — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 16, 2026

Diagram-based questions on Quantum Cryptography (BB84 Protocol)

Question 1 10.0 points

Examine the Bloch sphere representation of the BB84 protocol quantum states in the diagram below. Describe the position of each of the four states ($|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$) in terms of their polar and azimuthal angles on the Bloch sphere using rectilinear and diagonal bases.

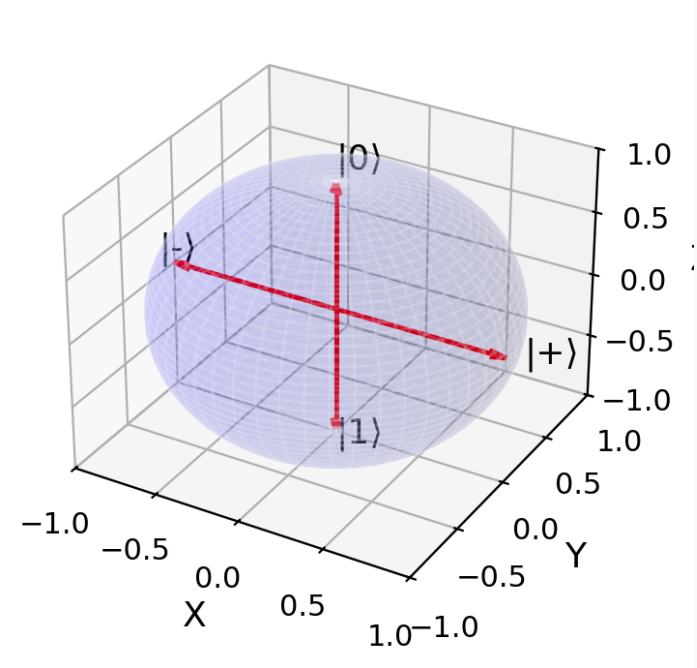

Fig. 1. A Bloch sphere diagram showing the quantum states for the BB84 protocol. The sphere should be centered with axes labeled as x, y, and z. The state |0⟩ should be at the north pole (z=+1), |1⟩ at the south pole (z=-1), |+⟩ on the equator along the x-axis, and |-⟩ on the equator along the negative x-axis. Include polar and azimuthal angles to illustrate the positions of these states in both rectilinear and diagonal bases. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • base editing window The diagram will FAIL review if any of these appear.

Question 2 10.0 points

Interpret the key exchange sequence in the diagram below for the BB84 protocol between Alice and Bob, including an intercept-resend attack by Eve. Explain how the protocol detects eavesdropping.

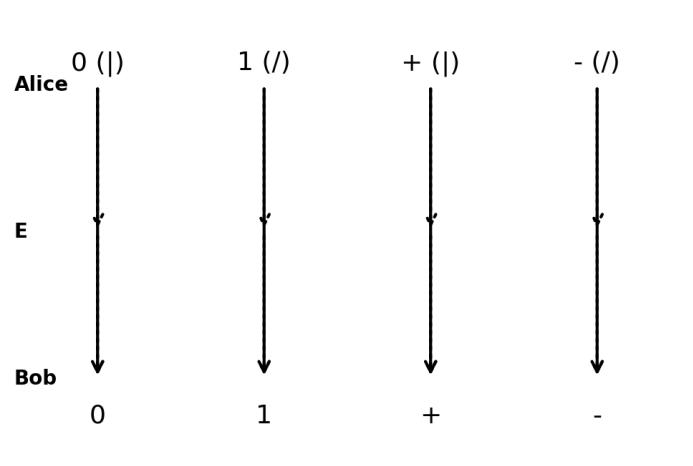

Fig. 2. A BB84 protocol sequence diagram showing the key exchange between Alice and Bob, including an intercept-resend attack by Eve. The diagram should include: 1) Alice preparing qubits in random bases (rectilinear or diagonal) and sending them to Bob. 2) Bob measuring the qubits in random bases and recording the results. 3) Eve intercepting the qubits, measuring them in random bases, and resending them to Bob. 4) The detection of eavesdropping through mismatched bases and error rates. Label each participant (Alice, Bob, Eve) and indicate the qubit states and bases used at each step. Include arrows to show the flow of qubits and highlight discrepancies caused by Eve's interference. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • If Eve • Eve • base editing window • syndrome table • eavesdropping detection • eavesdropping • logical qubit encoding path The diagram will FAIL review if any of these appear.

Question 3

10.0 points

Examine the quantum key distribution channel in the diagram below, which illustrates the probability of eavesdropping detection versus sifted key error rate. Discuss the trade-off between security and error rate.

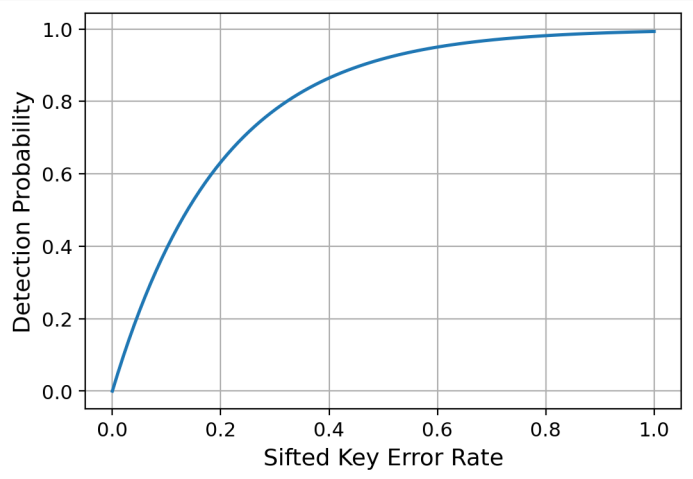

Fig. 3. A plot showing the probability of eavesdropping detection on the y-axis versus the sifted key error rate on the x-axis. The curve should illustrate the trade-off between security (probability of detection) and error rate, with labels for both axes and a title indicating the relationship being depicted. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • trade-off between security • eavesdropping detection • eavesdropping The diagram will FAIL review if any of these appear.

Question 4

10.0 points

Evaluate the quantum circuit shown below that demonstrates the no-cloning theorem. Explain how this circuit prevents the cloning of an arbitrary quantum state.

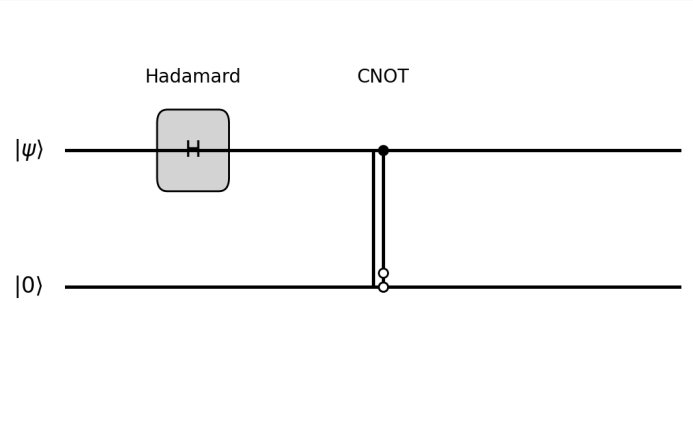

Fig. 4. A quantum circuit illustrating the no-cloning theorem. The circuit includes two qubit lines: one for the original quantum state |ψ⟩ and another for the ancilla qubit initialized to |0⟩. The circuit features a CNOT gate with the original qubit as the control and the ancilla as the target. Additionally, a Hadamard gate is applied to the original qubit before the CNOT gate. The diagram should clearly label the qubit lines and gates to show the circuit's structure.

Question 5 10.0 points

Interpret the correlation curve between sifted key error rate and eavesdropping probability in the quantum key distribution session as shown in the diagram below. How can this curve guide secure communication practices?

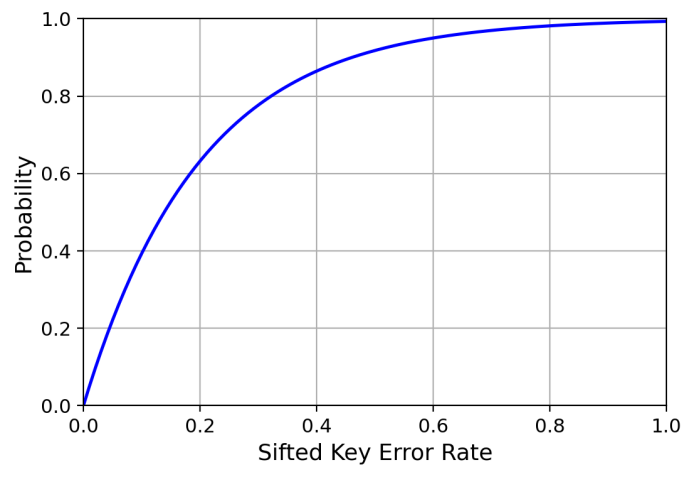

Fig. 5. A correlation curve showing the relationship between sifted key error rate and eavesdropping probability in a quantum key distribution session. The x-axis represents the sifted key error rate, and the y-axis represents the eavesdropping probability. The curve should illustrate how changes in the error rate affect the probability of eavesdropping, highlighting key points where secure communication practices are impacted. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • eavesdropping detection • eavesdropping The diagram will FAIL review if any of these appear.